ASCE 41-23 Tier 1 Checklists

FIRM:	
PROJECT NAME:	
SEISMICITY LEVEL:	
PROJECT NUMBER:	
COMPLETED BY:	
DATE COMPLETED:	
REVIEWED BY:	
REVIEW DATE:	

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

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Project Name _____
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17.11IO Structural Checklist for Building Type C1: Concrete Moment Frames

Table 17-23. Immediate Occupancy Structural Checklist for Building Type C1.

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Very Low Seismicity							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REDUNDANCY: The number of lines of moment frames in each principal direction is greater than or equal to 2. The number of bays of moment frames in each line is greater than or equal to 3.	5.5.1.1	A.3.1.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	INTERFERING WALLS: All concrete and masonry infill walls placed in moment frames are isolated from structural elements.	5.5.2.1.1	A.3.1.2.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	COLUMN SHEAR STRESS CHECK: The shear stress in the concrete columns, calculated using the Quick Check procedure of Section 4.4.3.2 is less than the greater of 100 lb/in.^{2} (0.69 MPa) or $2\sqrt{f_c'}$.	5.5.2.1.4	A.3.1.4.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	COLUMN AXIAL STRESS CHECK: The axial stress caused by unfactored gravity loads in columns subjected to overturning demands is less than $0.13f_c'$. Alternatively, the axial stress caused by overturning forces alone, calculated using the Quick Check procedure of Section 4.4.3.6 is less than $0.30f_c'$.	5.5.2.1.3	A.3.1.4.2	
Connections							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	CONCRETE COLUMNS: All concrete columns are doweled into the foundation, and the dowels are able to develop the tensile capacity of reinforcement in columns of the seismic-force-resisting system.	5.7.3.1	A.5.3.2	

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Low Seismicity (Complete the Following Items in Addition to the Items for Very Low Seismicity)							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	FLAT SLAB FRAMES: The seismic-force-resisting system is not a frame consisting of columns and a flat slab or plate without beams.	5.5.2.3.1	A.3.1.4.3	

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C	NC	N/A	U			
☐	☐	☐	☐	PRESTRESSED FRAME ELEMENTS: The seismic-force-resisting frames do not include any prestressed or posttensioned elements where the average prestress exceeds the lesser of 700 lb/in.^{2} (4.83 MPa) or $f_c^{'}/6$ at potential hinge locations. The average prestress is calculated in accordance with the Quick Check procedure of Section 4.4.3.8.	5.5.2.3.2	A.3.1.4.4
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	CAPTIVE COLUMNS: There are no columns at a level with height-to-depth ratios less than 75% of the nominal height-to-depth ratio of the typical columns at that level.	5.5.2.3.3	A.3.1.4.5
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	NO SHEAR FAILURES: The shear capacity of frame members is able to develop the moment capacity at the ends of the members.	5.5.2.3.4	A.3.1.4.6
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	STRONG COLUMN—WEAK BEAM: The sum of the moment capacity of the columns is 20% greater than that of the beams at frame joints.	5.5.2.1.5	A.3.1.4.7
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	BEAM BARS: At least two longitudinal top and two longitudinal bottom bars extend continuously throughout the length of each frame beam. At least 25% of the longitudinal bars provided at the joints for either positive or negative moment are continuous throughout the length of the members.	5.5.2.3.5	A.3.1.4.8
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	COLUMN-BAR SPLICES: All column-bar lap splice lengths are greater than $50d_b$ and are enclosed by ties spaced at or less than $8d_b$. Alternatively, column bars are spliced with mechanical couplers with a capacity of at least 1.25 times the nominal yield strength of the spliced bar.	5.5.2.3.6	A.3.1.4.9
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	BEAM-BAR SPLICES: The lap splices or mechanical couplers for longitudinal beam reinforcing are not located within $l_b/4$ of the joints and are not located in the vicinity of potential plastic hinge locations.	5.5.2.3.6	A.3.1.4.10
C	**NC**	**N/A**	**U**			
☐	☐	☐	☐	COLUMN-TIE SPACING: Frame columns have ties spaced at or less than $d/4$ throughout their length and at or less than $8d_b$ at all potential plastic hinge locations.	5.5.2.3.7	A.3.1.4.11

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C	NC	N/A	U			
☐	☐	☐	☐	STIRRUP SPACING: All beams have stirrups spaced at or less than $d/2$ throughout their length. At potential plastic hinge locations, stirrups are spaced at or less than the minimum of $8d_b$ or $d/4$.	5.5.2.3.7	A.3.1.4.12
C	**NC**	**N/A**	**U**	JOINT TRANSVERSE REINFORCING: Beam–column joints have ties spaced at or less than $8d_b$.	5.5.2.3.8	A.3.1.4.13
☐	☐	☐	☐			
C	**NC**	**N/A**	**U**	JOINT ECCENTRICITY: There are no eccentricities larger than 20% of the smallest column plan dimension between girder and column centerlines.	5.5.2.3.9	A.3.1.4.14
☐	☐	☐	☐			
C	**NC**	**N/A**	**U**	STIRRUP AND TIE HOOKS: The beam stirrups and column ties are anchored into the member cores with hooks of 135 degrees or more.	5.5.2.3.10	A.3.1.4.15
☐	☐	☐	☐			
C	**NC**	**N/A**	**U**	DEFLECTION COMPATIBILITY: Secondary components have the shear capacity to develop the flexural strength of the components and are Compliant with the following items in this table: COLUMN-BAR SPLICES, BEAM-BAR SPLICES, COLUMN-TIE SPACING, STIRRUP SPACING, and STIRRUP AND TIE HOOKS.	5.5.2.5.2	A.3.1.6.2
☐	☐	☐	☐			
C	**NC**	**N/A**	**U**	FLAT SLABS: Flat slabs or plates not part of the seismic-force-resisting system have continuous bottom steel through the column joints.	5.5.2.5.3	A.3.1.6.3
☐	☐	☐	☐			
Diaphragms						
C	**NC**	**N/A**	**U**	DIAPHRAGM CONTINUITY: Floor and roof diaphragms do not have expansion joints or vertical offsets, such as split levels, sawtooth, or clerestory configurations.	5.6.1.1	A.4.1.1
☐	☐	☐	☐			
C	**NC**	**N/A**	**U**	PLAN IRREGULARITIES: There is tensile capacity to develop the strength of the diaphragm at reentrant corners or other locations of plan irregularities.	5.6.1.4	A.4.1.7
☐	☐	☐	☐			
C	**NC**	**N/A**	**U**	DIAPHRAGM REINFORCEMENT AT OPENINGS: There is reinforcing around all diaphragm openings larger than 50% of the building width in either major plan dimension.	5.6.1.5	A.4.1.8
☐	☐	☐	☐			
Connections						
C	**NC**	**N/A**	**U**	UPLIFT AT PILE CAPS: Pile caps have top reinforcement, and piles are anchored to the pile caps; the pile cap reinforcement and pile anchorage are able to develop the tensile capacity of the piles.	5.7.3.5	A.5.3.8
☐	☐	☐	☐			

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Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Moderate and High Seismicity (Complete the Following Items in Addition to the Items for Very Low and Low Seismicity)							
Foundation System							
C	**NC**	**N/A**	**U**	OVERTURNING: The ratio of the least horizontal dimension of the seismic-force-resisting system at the foundation level to the building height (base/height) is greater than $0.6S_a$.	5.4.3.3	A.6.2.1	

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